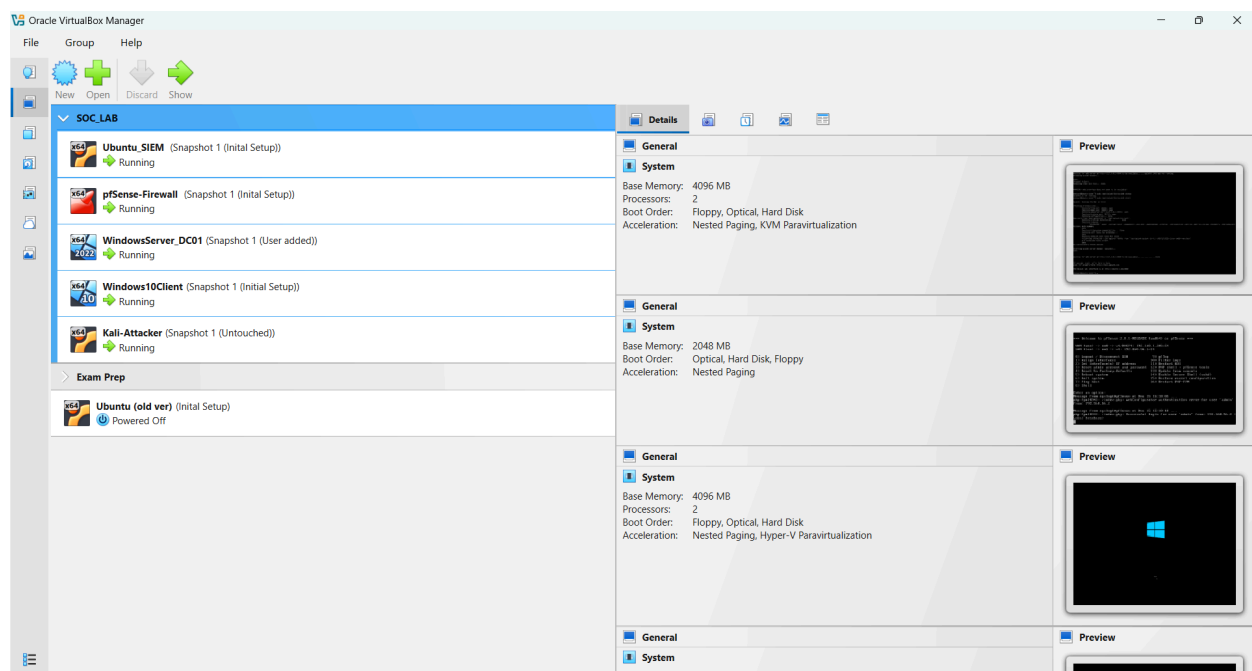


I built this SOC Home Lab as a way to learn cybersecurity through real, hands-on experience rather than just theory. I wanted to understand how different parts of a network interact, how attackers behave once they're inside an environment, and how defenders detect and investigate suspicious activity. Creating a small but realistic setup allowed me to explore traffic flows, logs, alerts, and security tools in a safe and controlled way. My goal throughout this project was to gain a practical understanding of how attacks unfold and how security operations teams respond to them.

To begin, I designed a simple architecture that included a firewall, a domain controller, a workstation, a SIEM, and an attacker machine. Everything ran inside VirtualBox so I could keep the environment isolated and easily reset it whenever needed.

Figure 1: VirtualBox environment showing all virtual machines used in the SOC Home Lab.



pfSense acted as the firewall and router for the network, giving me visibility into inbound and outbound traffic. Windows Server 2022 served as the Active Directory domain controller, handling user authentication and DNS. A Windows 10 workstation joined to this domain represented a typical user machine. For monitoring and analysis, I installed Splunk Enterprise on an Ubuntu server, and I used Sysmon on the Windows endpoint to capture detailed telemetry. Finally, Kali Linux acted as my attacker machine, allowing me to safely simulate reconnaissance, exploitation attempts, and authentication attacks.

Setting up pfSense was my first step because it became the gateway for the entire environment. I configured both a WAN and LAN interface and enabled DHCP on the internal network so my virtual machines could automatically receive IP addresses. I created simple firewall rules and

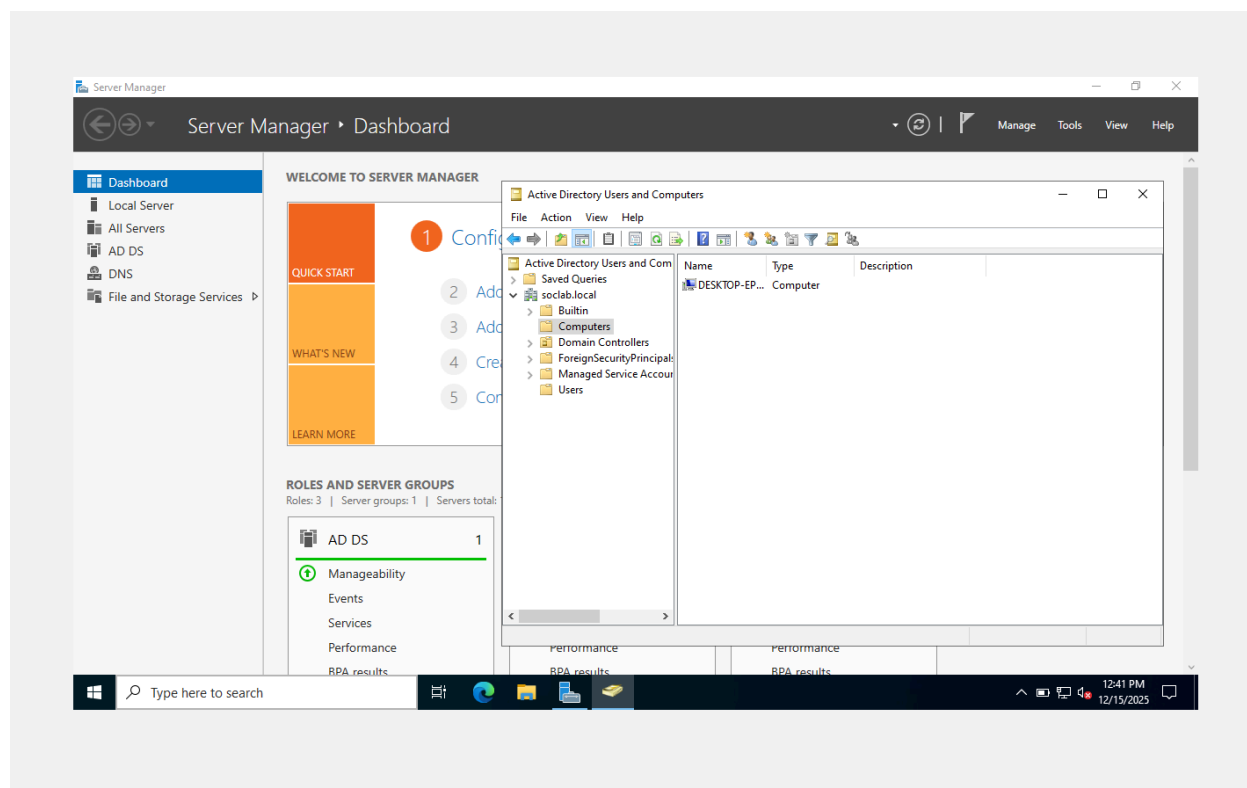
enabled logging so I could watch how the firewall responded when traffic from my attacker machine probed or scanned the network. Even basic actions like pinging or scanning from Kali immediately appeared in the pfSense logs, which was my first glimpse into the kinds of traces attackers leave behind.

Figure 2: pfSense firewall logs showing blocked ICMP traffic originating from the Kali attacker.



After that, I built the Active Directory domain. I installed Windows Server 2022, promoted it to a domain controller, and created the domain “soclab.local.” Inside the domain, I created user accounts and simple organizational units.

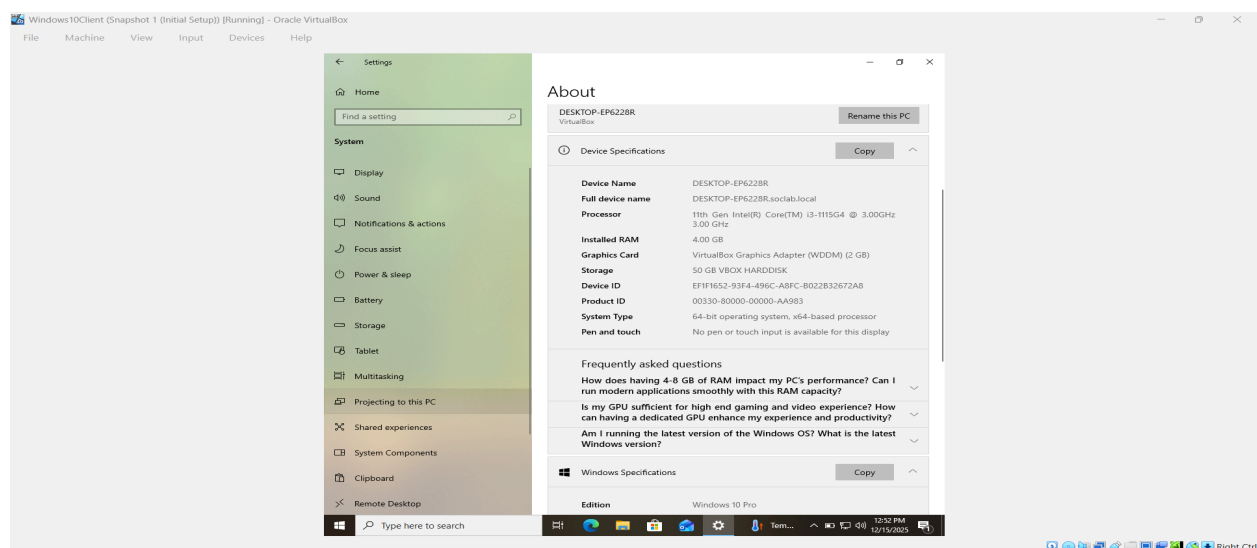
Figure 3: Active Directory Users and Computers showing the soclab.local domain



Working with Active Directory helped me understand how authentication works in a real environment and why domain controllers generate some of the most valuable logs for incident response. Every login attempt—successful or not—became part of the story that Splunk later helped me analyze.

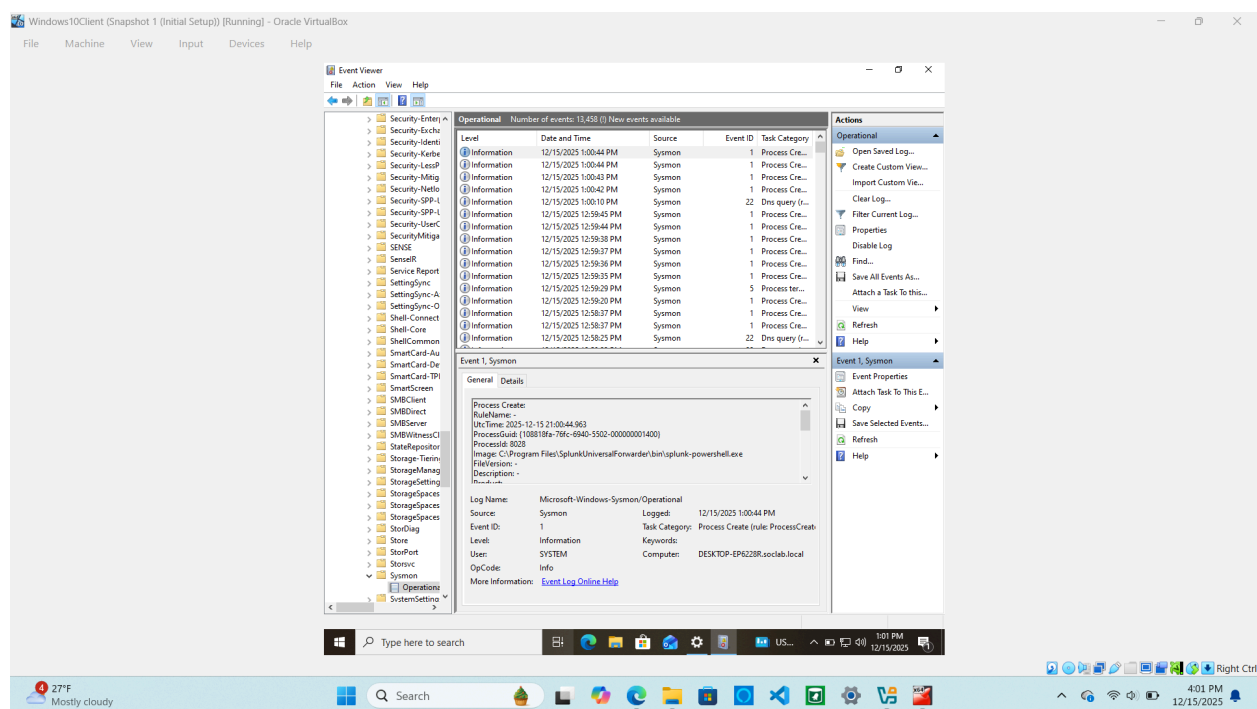
Once the domain was ready, I configured the Windows 10 workstation and joined it to the Active Directory domain. This workstation served as a realistic endpoint where user activity, authentication events, and system behavior could be observed and analyzed.

Figure 4: Windows 10 workstation successfully joined to the Active Directory domain



After confirming the workstation was properly integrated into the domain, I installed Sysmon to collect detailed endpoint telemetry. Sysmon allowed me to capture process creation, network connections, and other low-level endpoint activity that attackers often generate during an intrusion.

Figure 5: Sysmon operational logs visible in Windows Event Viewer on the domain-joined workstation



To centralize and analyze these events, I installed the Splunk Universal Forwarder on the Windows workstation and configured it to send Sysmon logs to my Splunk Enterprise instance. This allowed endpoint activity to be viewed alongside other security data in a single location.

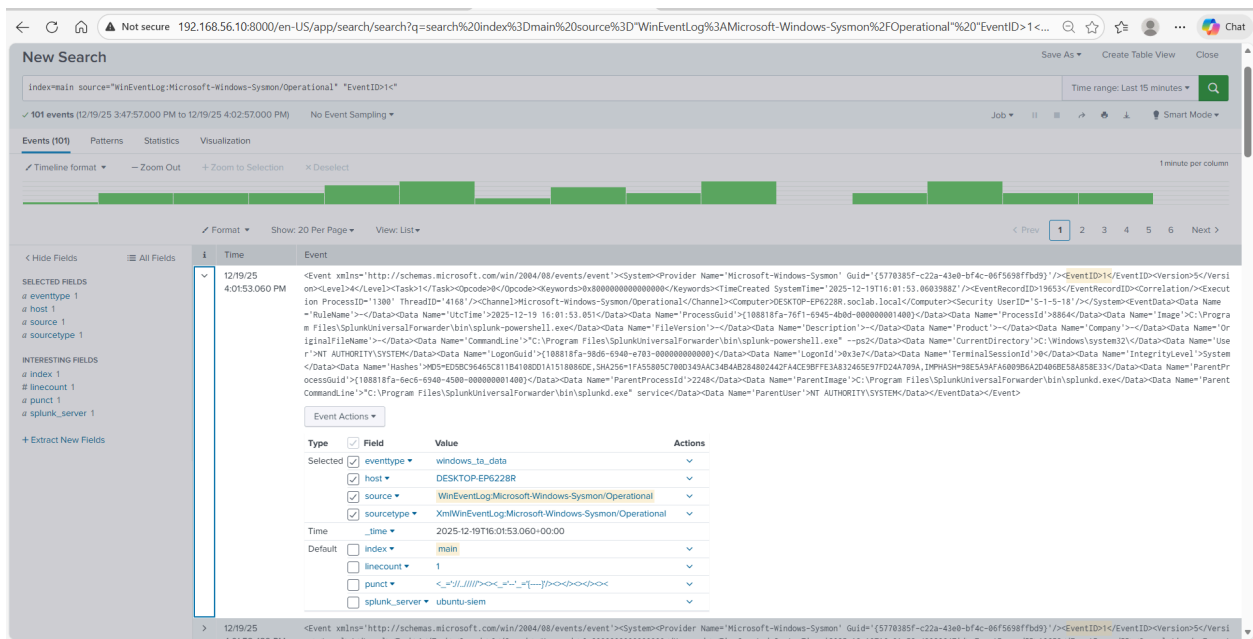
Figure 6: Sysmon process creation events from the Windows workstation successfully ingested into Splunk

The screenshot displays the Splunk Enterprise search interface. The search bar contains the query: `Index=* sourcetype=XmlWinEventLog:Microsoft-Windows-Sysmon/Operational`. The search results show 983 events from 12/14/25 9:00:00.000 PM to 12/15/25 9:07:27.000 PM. The interface is set to 'Timeline format' and 'Zoom Out'. The search results table shows the following data:

Time	Event
12/15/25 9:06:26.265 PM	<Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'><System><Provider Name='Microsoft-Windows-Sysmon' Guid='{5770385f-c22a-43e0-bf4c-86f5698fbd9}' /><EventID>1</EventID><Version>5</Version><Level>4</Level><Task>1</Task><Opcode>0</Opcode><Keywords>0x8000000000000000</Keywords><Time Created>SystemTime='2025-12-15T21:06:26.2652400Z' /><EventRecordID>13506</EventRecordID><Correlation><Execution ProcessID='1300' ThreadID='4168' /><Channel>Microsoft-Windows-Sysmon/Operational</Channel><Computer>DESKTOP-EP6228R.soclab.local</Computer><Security UserID='S-1-5-18' /></System><EventData><Data Name='RuleName'></Data><Data Name='UtcTime'>2025-12-15 21:06:26.259</Data><Data Name='ProcessGuid'>{108818fa-7852-6940-7f02-000000001400}</Data><Data Name='ProcessId'>7304</Data><Data Name='Image'>C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe</Data><Data Name='FileVersion'></Data><Data Name='Description'></Data><Data Name='Product'></Data><Data Name='Company'></Data><Data Name='OriginalFileName'></Data><Data Name='CommandLine'>'C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe' --ps2</Data><Data Name='CurrentDirectory'>C:\Windows\system32</Data><Data Name='User'>NT AUTHORITY\SYSTEM</Data><Data Name='LogonGuid'>{108818fa-98d6-6940-e703-000000000000}</Data><Data Name='LogonId'>0x3e7</Data><Data Name='TerminalSessionId'>0</Data><Data Name='IntegrityLevel'>System</Data><Data Name='Hashes'>MD5=ED5BC96465C811B4108D01A1518086DE, SHA256=1FA55805C700D349AAC3484AB284802442FA4CE9BFE3A832465E97FD24A709A, IMPHASH=98E5A9AF6009B6A2D406BE58A85E33</Data><Data Name='ParentProcessGuid'>{108818fa-6ec6-6940-4500-000000001400}</Data><Data Name='ParentProcessId'>2248</Data><Data Name='ParentImage'>C:\Program Files\SplunkUniversalForwarder\bin\splunkd.exe</Data><Data Name='ParentCommandLine'>'C:\Program Files\SplunkUniversalForwarder\bin\splunkd.exe' service</Data><Data Name='ParentUser'>NT AUTHORITY\SYSTEM</Data></EventData></Event>

Installing Splunk Enterprise on Ubuntu tied the entire environment together. I created indexes for Windows logs and Sysmon logs and configured Splunk to receive data from the forwarder. Once the logs started flowing in, Splunk became the central view of everything happening inside the lab.

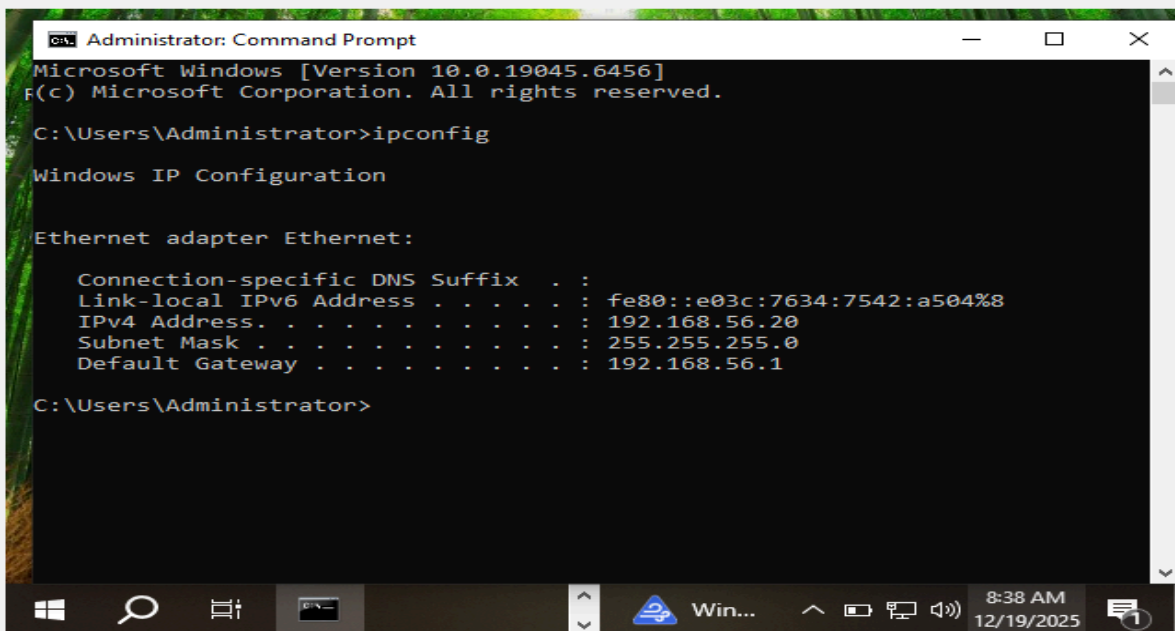
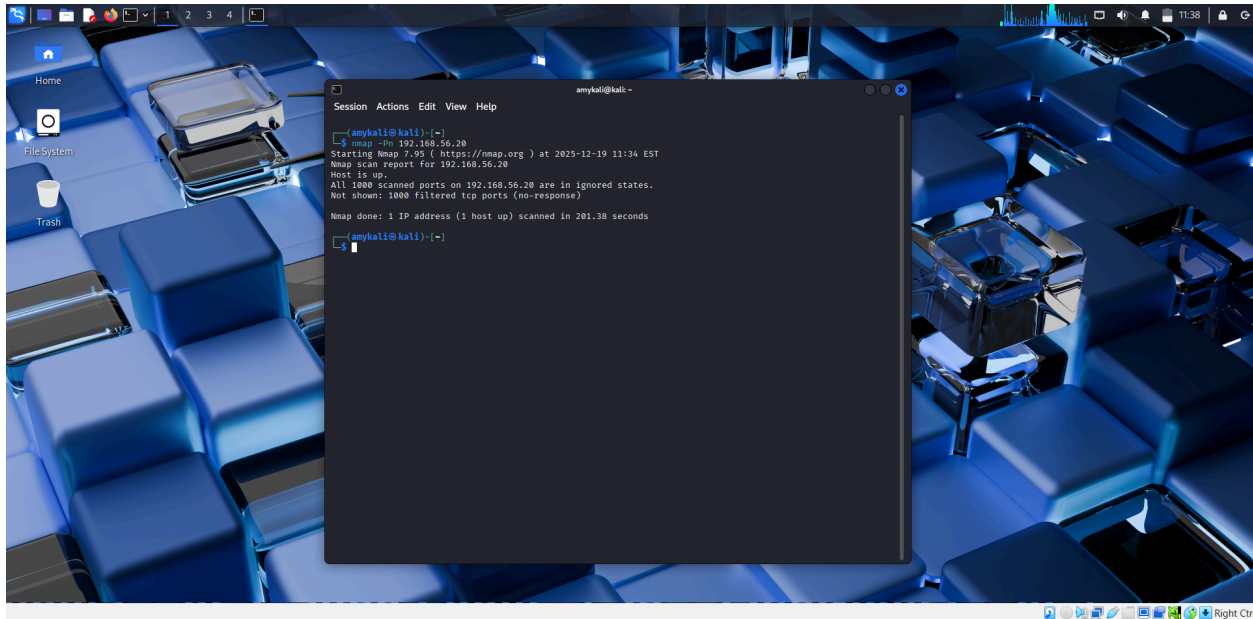
Figure 7: Splunk search showing Sysmon Operational logs with an expanded process creation event (Event ID 1):



I ran searches, built simple dashboards, and learned how to identify patterns—like repeated login failures, unusual process creation activity, and connections coming from unexpected locations. This helped me understand the workflow of a SOC analyst: collect data, analyze events, investigate patterns, and determine whether something is normal or suspicious.

With the defensive environment ready, I turned my attention to Kali Linux to simulate attacker behavior. I began with simple reconnaissance by scanning the network using nmap and other enumeration tools.

Figure 8: Kali Linux performing an Nmap scan against the Windows 10 client (192.168.56.20). The host was detected as online, but all scanned ports were filtered, indicating a hardened endpoint configuration.



Even though the host did not respond to ICMP probes, forcing host discovery with -Pn confirmed the system was online, demonstrating how attackers adapt when basic reconnaissance is blocked. These scans immediately appeared in pfSense logs, Sysmon logs, and Splunk dashboards.

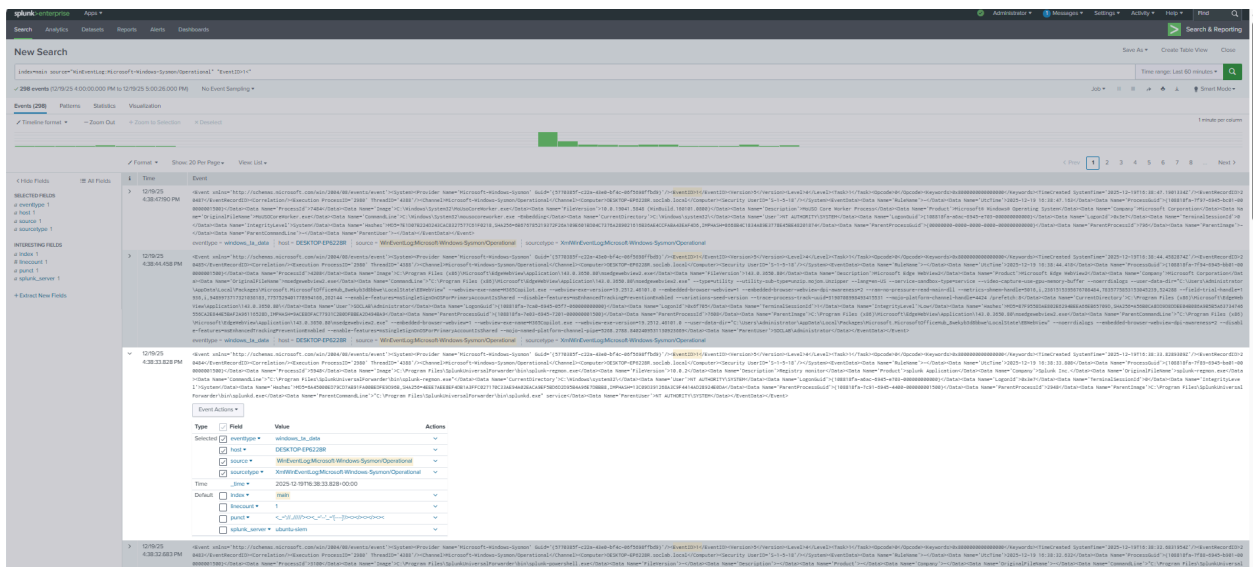
Figure 9: Sysmon process creation events ingested into Splunk, confirming that reconnaissance activity against the Windows 10 client was detected and centrally logged.

```
(amykali@kali)-[~]
$ nmap -sT -p 445,3389 -Pn 192.168.56.20
Starting Nmap 7.95 ( https://nmap.org ) at 2025-12-19 11:46 EST
Nmap scan report for 192.168.56.20
Host is up.

PORT      STATE SERVICE
445/tcp    filtered microsoft-ds
3389/tcp    filtered ms-wbt-server

Nmap done: 1 IP address (1 host up) scanned in 3.08 seconds

(amykali@kali)-[~]
```



I then attempted authentication attacks, such as password guessing and brute-force attempts against the domain controller. These actions generated failed login events, process execution logs, and network activity that all became visible inside Splunk. Observing how my actions in Kali produced recognizable patterns across the environment helped me connect theory to practice. It showed me how even small attacker actions create a footprint and how important it is for security teams to have proper logging and monitoring in place.

Throughout this process, I learned how to interpret Windows event logs, how Sysmon reveals detailed endpoint activity, and how Splunk makes it possible to explore and correlate these events. I gained practical experience with Active Directory, firewall behavior, network segmentation, detection queries, and attack analysis. Most importantly, I learned how a SOC environment “feels” during normal use versus during an attack, which is something no textbook can fully capture.

Working on this lab gave me a deeper appreciation for how defenders think and how attackers operate. It also helped me build confidence in my ability to analyze real security events and understand the story behind them. Going forward, I plan to expand this lab by adding tools like Wazuh or Suricata, incorporating Linux servers with auditd logging, building more Splunk dashboards, and experimenting with automated detection rules. This project is something I will

continue to grow as I grow in cybersecurity, and each new component will help me develop stronger blue-team skills.